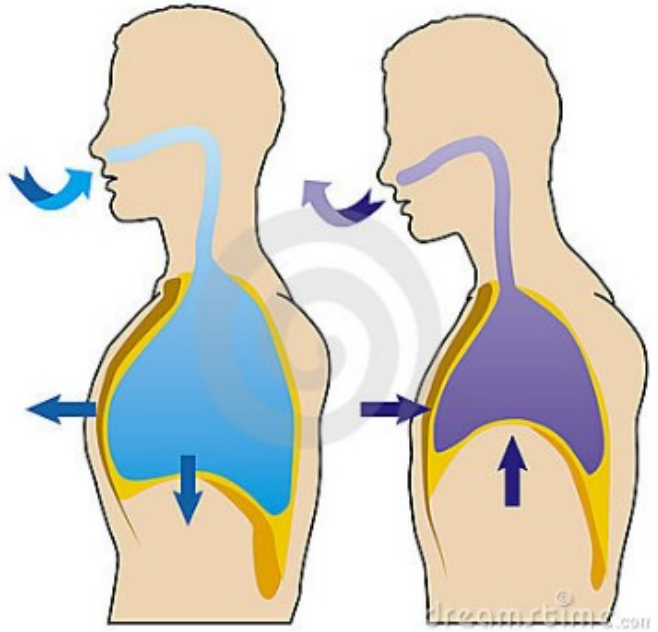
Regulation of Gas Concentrations

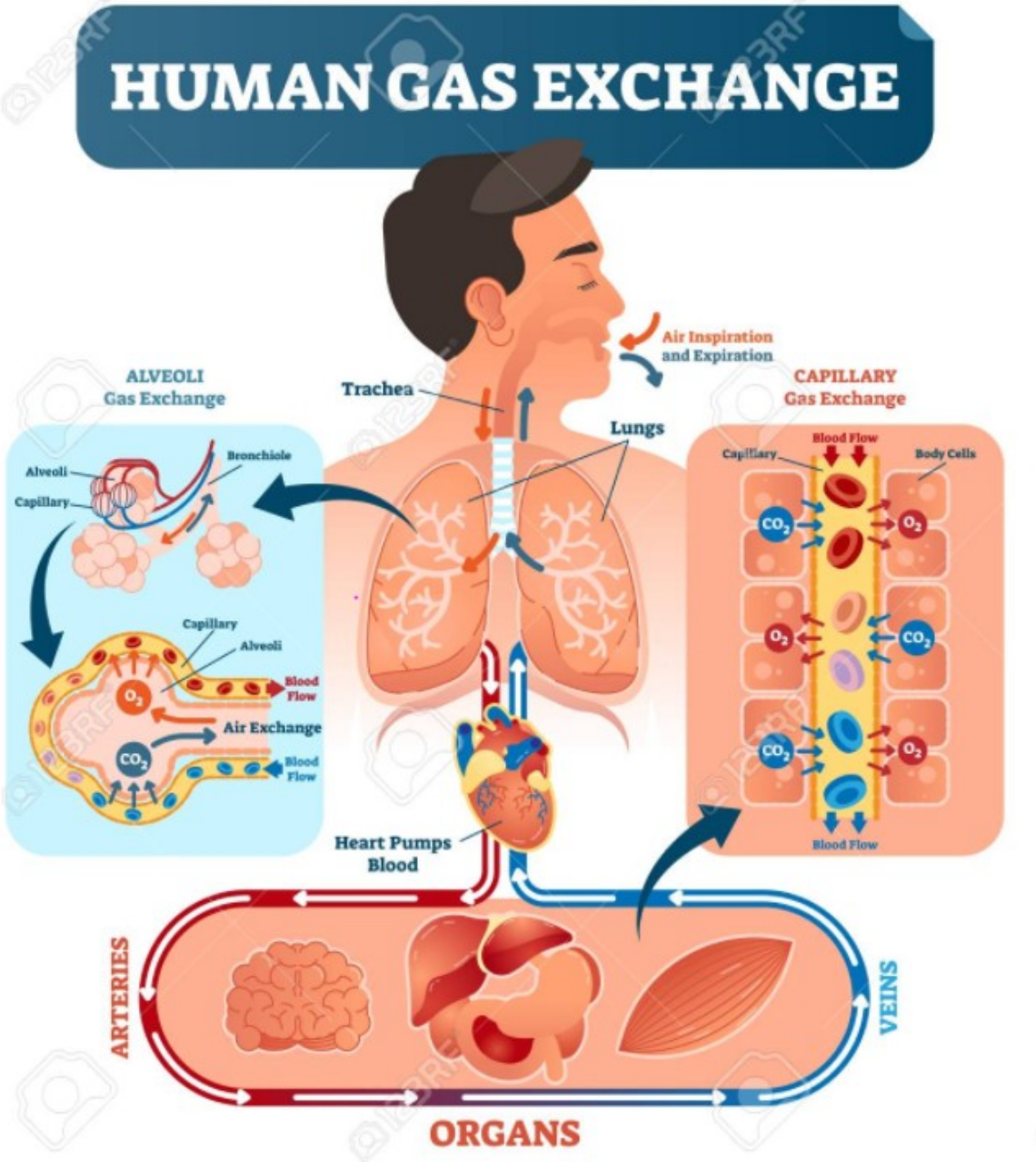


Overview

- For homeostasis, cells need:
  - Continuous supply of oxygen (used in cellular respiration)
  - Ability to excrete carbon dioxide waste
- Levels of O_2 and CO_2 in the blood must be regulated.
- Respiratory and Circulatory systems work together to do this.

- Respiratory system allows:
  - Intake of O_2 from environment
  - Excretion of CO_2 from body

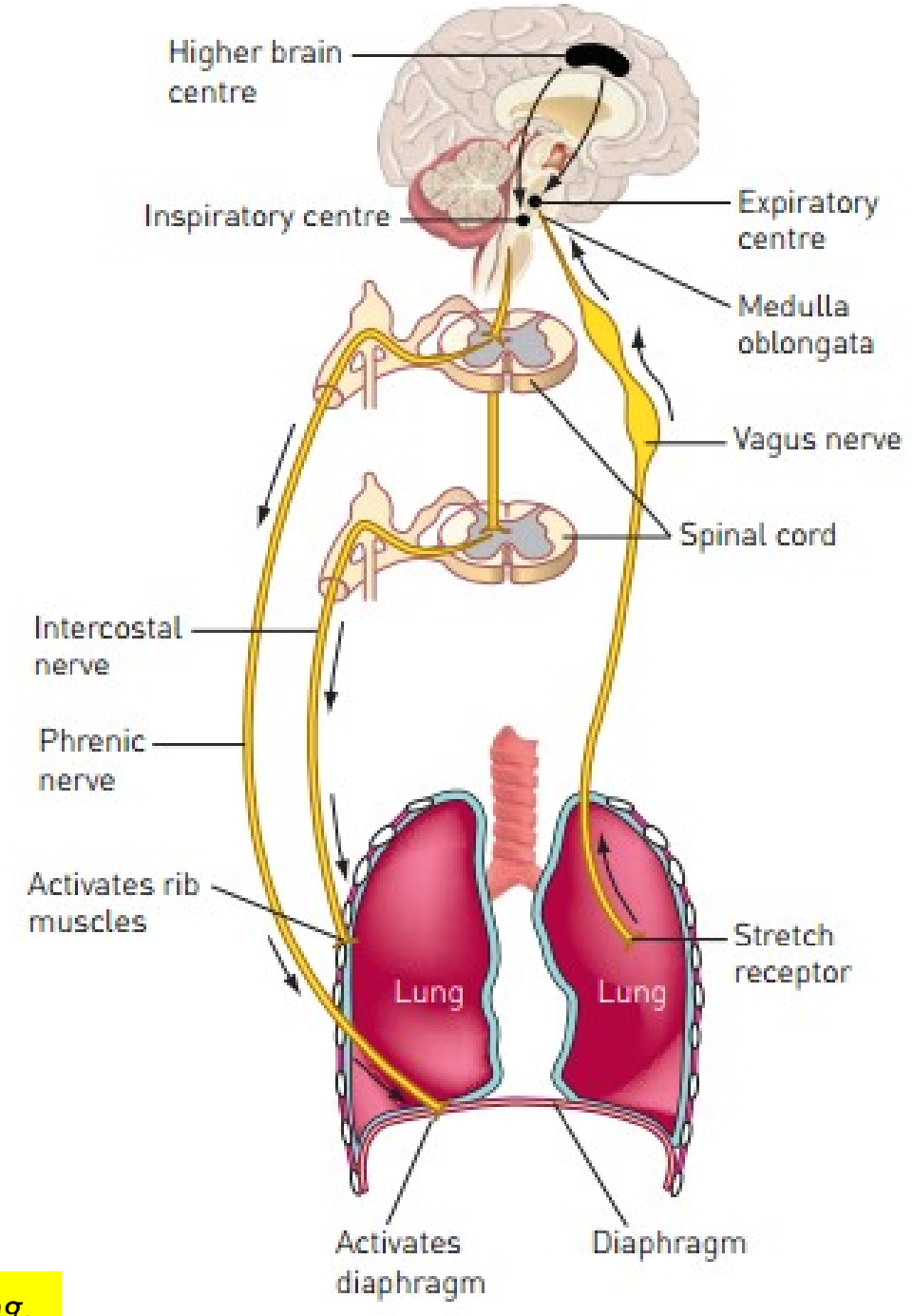
- Cardiovascular/Circulatory System
  - Allows O_2 to be moved from lungs to cells
  - Allows CO_2 to be moved from cells to lungs and excreted



Learning Aim: Summarise how gas exchange occurs in the body between environment, bloodstream and cells.

Mechanics of Breathing

- Breathing uses muscles:
  - Diaphragm
  - Intercostal muscles
- Use of these muscles needs nerve stimulation
  - Diaphragm: Phrenic nerve
  - Intercostal muscles: Intercostal nerves
- Spinal and Intercostal nerves
  - originate from spinal cord at neck and thorax
  - Controlled by respiratory centre in medulla oblongata
    - Has expiratory centre and inspiratory centre



Learning Aim: Describe the nerve and muscle pathways involved in breathing.

Control of Rate of Breathing (Respiratory Rate)

- O_2 and CO_2 carried in blood, concentrations affect breathing rate.
- Respiratory Rate is mostly controlled by CO_2 concentration in plasma:



Note: Concentration of CO_2 affects H^+ concentration. As CO_2 levels in the blood rise, more carbonic acid is produced, so more H^+ becomes dissolved in the blood plasma, and the pH drops.

***Remember*: rising CO_2 levels are the stimulus for breathing**, not falling O_2 levels, except in extreme circumstances.

Learning Aim: Discuss how CO_2 concentration impacts blood pH

Stimulus for Blood Gas Homeostasis: Carbon Dioxide Concentration

- CO_2 that enters the bloodstream is carried in plasma:
  - as dissolved CO_2 (approx. 10 %)
  - as dissolved carbonic acid: $H^+ + HCO_3^-$ (approx. 80 %)
  - a small amount is bound to RBC (10 %) but is not important in gas homeostasis
- CO_2 is the major stimulus in regulation of gas homeostasis via respiratory rate.
- Small rise in blood CO_2 → big difference in rate of breathing.
- Can be detected as dissolved CO_2 or as concentration of H+ (pH).

NOTE: ↑ CO_2 leads to ↑ H+ which means ↓ pH (low pH means more acidic)

Learning Aim: Understand that CO_2 concentration and its effect on pH are the major stimulus for regulation of respiratory rate.

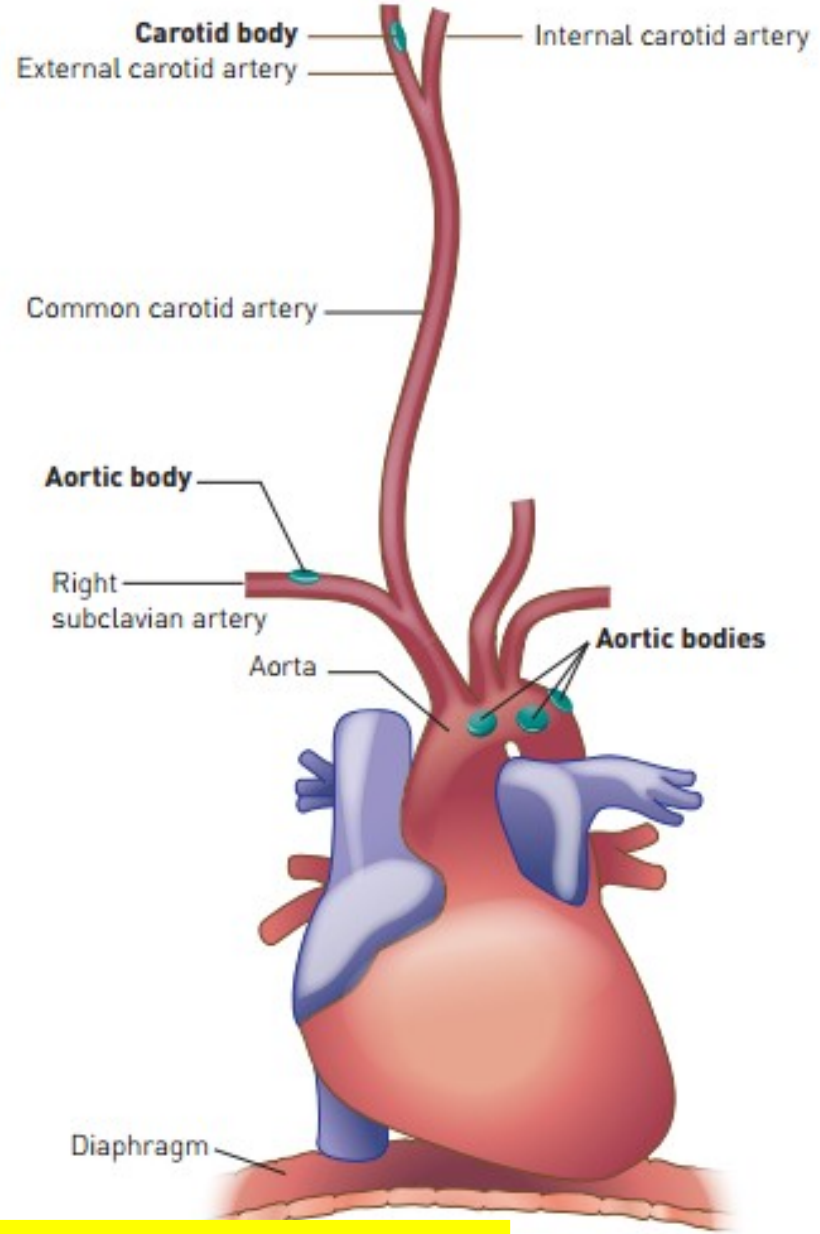
Receptors for regulation of blood gas concentration

Blood gas concentration is detected by ***chemoreceptors*** in the aorta and carotid artery. These then send signals to the medulla oblongata in the brain.

- Chemoreceptors in the aortic body detect CO_2 and O_2 concentration
- Chemoreceptors in the carotid body detect CO_2 concentration, H+ concentration (pH) and O_2 concentration

The medulla oblongata also has ***chemoreceptors*** that directly detect blood gas concentrations.

- Chemoreceptors in the medulla oblongata detect H+ concentration (blood pH).



Learning Aim: Identify the major receptors and their locations for blood gas concentration and what each detects.

Homeostasis when CO_2 Levels Rise

Stimulus

Increased blood plasma CO_2 levels (and drop in pH—more acidic) due to: decreased excretion from breathing (decreased respiratory rate and depth), or increased production due to increased cellular respiration (for example due to exercise)



Receptor

Chemoreceptors in aortic arch and carotid (peripheral chemoreceptors) detect rising CO_2

Chemoreceptors in medulla oblongata (central chemoreceptors) detect falling pH (increasing acidity)



Message

Autonomic NS

Signalling within medulla oblongata

Control Centre / Modulator

Respiratory Centre in Medulla Oblongata

Message

Autonomic NS signalling (symp)

Note: CO_2 levels are the most important stimulus in controlling blood gas levels—O_2 is less important

Effector

Respiratory muscles

Response

Increased rate and depth of breathing allows increased CO_2 excretion.

$$CO_2 + H_2O \rightleftharpoons H_2CO_3 \rightleftharpoons H^+ + HCO_3^-$$

Carbon dioxide — Water — Carbonic acid — Hydrogen ion — Bicarbonate ion

Feedback

Negative Feedback: CO_2 levels and pH return to normal, stimulus ceases, cycle stops

Learning Aim: Discuss how homeostasis is maintained when CO_2 levels rise.

Homeostasis when CO_2 levels rise above set point

1: Stimulus
Rising blood CO_2 levels/rising H+ levels (falling pH)
Due to:
- Increased exertion and cellular respiration
- Decreased rate and depth of breathing

2: Receptors
a) Rising CO_2 levels detected by Chemoreceptors in the aortic body and carotid arch
b) Rising H+ levels (falling pH) detected by chemoreceptors in the carotid arch and medulla oblongata

Messenger:
Nervous System

3: Modulator/Control Centre

Respiratory centre in the medulla oblongata receives information and coordinates actions to reduce CO_2 concentration

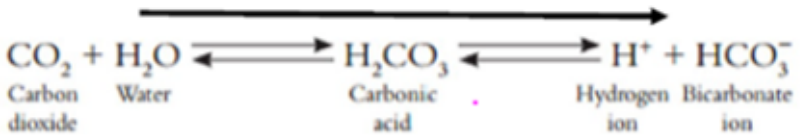
Messenger:
Autonomic Nervous System

4: Effectors
Diaphragm
Intercostal Muscles

5: Response
Diaphragm and Intercostal muscles act to increase rate and depth of breathing.

More CO_2 is excreted via the respiratory tract, so blood CO_2 falls, H+ concentration falls (pH rises)

6: Feedback
Negative feedback:
CO_2/H+/pH levels return to set point
Stimulus Ceases
Cycle Stops



Learning Aim: Draw a detailed, annotated feedback loop (steady state control model) to show how homeostasis is maintained when CO_2 levels rise.

Homeostasis when CO_2 Levels Fall

Stimulus

Decreased blood plasma CO_2 levels (and rise in pH—more alkaline) due to: increased excretion from breathing (eg hyperventilation which may be voluntary, or due to pain, or other factors)

Receptor

Chemoreceptors in aortic arch and carotid (peripheral chemoreceptors) detect falling CO_2

Chemoreceptors in medulla oblongata (central chemoreceptors) detect rising pH (decreasing acidity)

Message

Autonomic NS

Signalling within medulla oblongata

Control Centre / Modulator

Respiratory Centre in Medulla Oblongata



Message

Autonomic NS signalling

Effector

Respiratory muscles (inhibition)

Response

Decreased rate and depth of breathing allows decreased-CO_2 excretion.

$$CO_2 + H_2O \rightleftharpoons H_2CO_3 \rightleftharpoons H^+ + HCO_3^-$$

Carbon dioxide — Water — Carbonic acid — Hydrogen ion — Bicarbonate ion

Feedback

Negative Feedback: CO_2 levels and pH return to normal, stimulus ceases, cycle stops

Learning Aim: Discuss how homeostasis is maintained when CO_2 levels fall.

Homeostasis when CO_2 levels fall below set point

1: Stimulus
Falling blood CO_2 levels/falling H+ levels (rising pH)
Due to:
- Hyperventilation (due to pain, anxiety, or voluntary choice)

2: Receptors
a) Falling CO_2 levels detected by Chemoreceptors in the aortic body and carotid arch
b) Falling H+ levels (rising pH) detected by chemoreceptors in the carotid arch and medulla oblongata

Messenger:
Nervous System

3: Modulator/Control Centre

Respiratory centre in the medulla oblongata receives information and coordinates actions to increase CO_2 concentration

Messenger:
Autonomic Nervous System

4: Effectors
Diaphragm
Intercostal Muscles

5: Response
Diaphragm and Intercostal muscles act to decrease rate and depth of breathing.

Less CO_2 is excreted via the respiratory tract, so blood CO_2 rises, H+ concentration rises (pH falls)

6: Feedback
Negative feedback:
CO_2/H+/pH levels return to set point
Stimulus Ceases
Cycle Stops

$$CO_2 + H_2O \rightleftharpoons H_2CO_3 \rightleftharpoons H^+ + HCO_3^-$$

Carbon dioxide; Water; Carbonic acid; Hydrogen ion; Bicarbonate ion

Learning Aim: Draw a detailed, annotated feedback loop (steady state control model) to show how homeostasis is maintained when CO_2 levels fall.

Oxygen concentration and regulation of blood gas levels

- Concentration of O_2 in blood falls as it is consumed by cells.
- Falling O_2 concentration only has small effect on breathing rate.
- Plays little part in regulation of breathing, except if O2 concentration become EXTREMELY low:

1: Very low O_2 stimulates chemoreceptors:
   - Carotid and Aortic bodies
   - Medulla Oblongata

2: Nerve impulses transmitted to respiratory centre

3: Phrenic and Intercostal nerve stimulation increases breathing to restore O_2 levels.

- **This only occurs in extreme circumstances.** Day to day, CO_2 is the stimulus for increased respiratory rate.

Learning Aim: Discuss the role of O_2 concentration in regulating blood gas levels and respiratory rate.

Voluntary Control of Breathing

- Breathing can be voluntarily controlled – up to a point!
- Voluntary control of breathing is necessary for speech.
- Brain connections from cerebral cortex to spinal cord – bypasses respiratory centre in medulla oblongata.
- We can voluntarily hold our breath, or voluntarily hyperventilate.

- However: if we leave it too long:
  - Extreme Breath holding:
    - CO_2 levels in blood increase, blood becomes increasingly acidic.
    - Overwhelming urge to breathe
    - Combination of increasing acidity and falling O_2 causes unconsciousness and the medulla takes over.

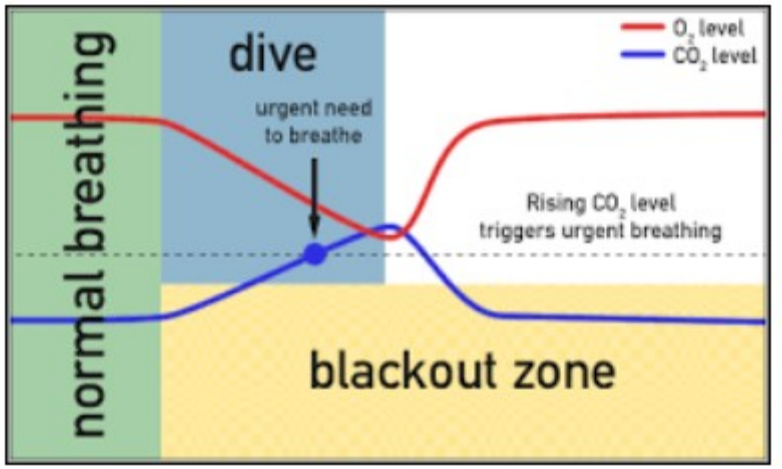
  - Extreme Voluntary hyperventilation:
    - CO_2 levels in blood decrease, blood becomes increasingly alkaline.
    - Dizziness, then unconsciousness and medulla takes over.

Learning Aim: Describe the nerve pathways involved in voluntary control of breathing, and why voluntary control is limited.

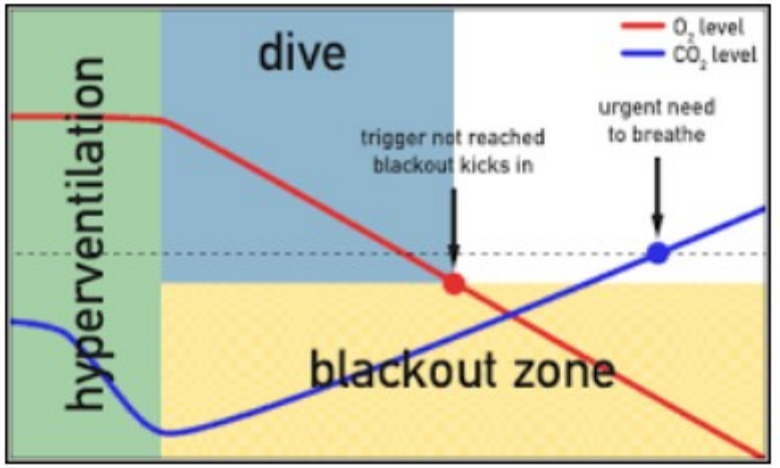
Shallow Water Blackout

the dangers of hyperventilation before swimming or diving

- Normal breathing:
  - Rising CO_2 levels trigger need to breathe.
  - This happens long before O_2 levels get low enough for blackout.



- Hyperventilation
  - Artificially lowers CO_2 levels
  - Need to breathe is suppressed.
  - Strong urge to breathe may not occur until *after* O_2 blackout
  - If you're underwater when that happens, you'll drown.



Learning Aim: Discuss the physiological mechanisms involved in shallow water blackout.

Exercise and Breathing – "Oxygen Debt"



- Muscle contraction during exercise consumes O_2 and produces CO_2 waste.
  - If not enough O_2 available during exercise, muscles use anaerobic respiration, and build up lactic acid.
  - After exercise, the lactic acid is put back through aerobic respiration: Oxygen used to do this, and CO_2 is produced as waste, which keeps CO_2 levels high after exercise is finished.
  - This is why we keep puffing after exercise is finished – our body is supplying the oxygen needed to process lactic acid, and getting rid of the CO_2 being produced as the lactic acid is processed.
- The same mechanisms involving chemoreceptors, medulla oblongata and nervous control of breathing apply: rate and depth of breathing increases to get rid of excess CO_2

Learning Aim: Discuss the physiological mechanisms involved in oxygen debt.